Jessica Vareed

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LinkedIn | GitHub

MSc Computer Science student specializing in Big Data Engineering, with hands-on experience building machine learning applications, data pipelines, backend APIs, and automation workflows. Experienced in Python, Java, REST APIs, SQL, Spark, Docker, Kubernetes, and CI/CD, with practical experience applying LLM-assisted development and model evaluation. Interested in building GenAI systems, that includes RAG, LLM agents, AI agents for automation and evaluation frameworks.

EDUCATION

- **Universiteit van Amsterdam & Vrije Universiteit Amsterdam, Amsterdam,** MSc Computer Science- Big Data Engineering Track (joint degree) (2025 - Expected June, 2027)
  Relevant modules: Data Engineering, Web Services, Data Mining, Data Systems Architecture
- **New Horizon College of Engineering, Bangalore,** BE Honours- Computer Science and Engineering
  CGPA - 9.5 /10 (2020 – 2024)

EXPERIENCE

- **Vrije Universiteit Amsterdam - Teaching assistantship: Digitization and Sustainability (Sep, 2026 - Oct 2026) ; Automata and Complexity (Feb, 2026 - Mar, 2026)-** Student support and conduct exercise-solving lab sessions, assisting with grading assignments, guiding with concept understanding and feedbacks.
- **Junglee Games- Software Development Engineer in Test (Aug, 2024 – Jul, 2025)-** Developed automated backend, frontend, and Rest API test workflows using Java, Selenium, TestNG, and RestAssured, achieving ~70% automation coverage. Integrated GitHub Copilot to generate and optimize test cases, and collaborated with developers in the Agile environment through CI/CD pipelines (Jenkins) for automated build and release workflows.
- **Junglee Games- Software Development Engineer in Test Intern (Aug, 2023 –Jul, 2024)-** Automated feature, regression, and peripheral workflows and validated payment gateways, SQL and NoSQL databases, and microservice interactions, improving testing efficiency by 40–50%.

PROJECTS

- **Dataset Robustness Profiling - Python | Scikit-Learn | Spark MLlib** - Built a Python/Spark framework to profile dataset quality and pipeline robustness through controlled perturbations, applying model evaluation techniques to quantify downstream ML degradation. Derived Dataset Robustness Vectors (DRV) to compare stability across text datasets using Logistic Regression, Naive Bayes, and SVM.
- **URL Shortener Web Service - Python | Flask | REST API | SQL | Docker** - Designed and implemented microservice based REST APIs for authentication and URL shortening service using Flask with persistent SQL-based storage and containerized deployment using Docker. Orchestrated the service via Kubernetes with persistent volume storage and NodePort-exposed REST endpoints following cloud-native deployment.
- **Analysing the Virality of Sustainability-Related Hashtags - Python | Matplotlib | Seaborn -** Collected and analyzed Mastodon & Bluesky data using APIs; built co-occurrence networks, engagement metrics, and user-activity models to study virality patterns in sustainability hashtags.

PUBLICATIONS

- **Machine Learning Predictive Models for Autism Spectrum Disorder Using Eye-Tracking Technology | IEEE, 2024-** Developed and implemented deep learning and machine learning models to process and analyze eye-tracking datasets for early detection of Autism Spectrum Disorder (ASD). Improved model reliability through advanced preprocessing, feature engineering, and hybrid model design. Increased model accuracy by 10% after expanding the dataset and introducing hybrid deep learning and classical ML architectures.
- **Prescription Recognition for the Visually Impaired using Convolutional Neural Networks (CNN) | IJSRCSEIT, 2023-** Designed and deployed a CNN-based image recognition system for automated prescription label reading and speech synthesis output, enabling visually impaired users to manage medications efficiently. Implemented end-to-end pipeline including image preprocessing, model training, inference, and integration with text-to-speech systems, showcasing practical application of computer vision and software engineering principles.

SKILLS

- **Programming**- Python, Java, C, JavaScript, React, Angular, SQL, Flask, REST APIs, Git, GitHub
- **AI, ML & Data-** Pandas, NumPy, Matplotlib, Feature Engineering, NLP, Model Evaluation, LLM, PySpark, Apache Spark MLlib, TensorFlow, scikit-learn, Data Visualization
- **Automation & Testing**- Selenium, Appium, TestNG, JUnit, RestAssured, Jira, Postman, API Automation
- **Cloud and Databases Management**-Docker, CI/CD pipelines, Kubernetes, ArgoCD, SQL (PostgreSQL), NoSQL (MongoDB), Redis, Kafka